

 UTHM Universiti Tun Hussein Onn Malaysia FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY	Course Code	BIT22103	Item	PROJECT
	Course	INTEGRATED SYSTEM TECHNOLOGIES	Date Release	W8 - MAY 2026
Title	Secure-by-Design E-Commerce Platform: API Integration, Data Conversion and Client-Server Architecture			
Objectives	At the end of this task, student should be able to: • Understand the concept of Client-Server Architecture and API-based communication. • Apply secure system design principles (authentication, authorization, data protection). • Demonstrate key skills in Backend API development and Frontend integration. • Demonstrate the use of External API integration within a system. • Perform data conversion techniques (JSON to CSV/XML). • Implement a secure communication flow between client and server			
Method	• System development (Backend API & Frontend) • Implementation of secure authentication and RBAC • Integration with external APIs • Testing and observation of system workflow • Books/Journals; Internet			
Mode	• Group (4 - 5 members only)			
Submission	• Report - Week 13 • Presentation - Week 14 - 15 (will be scheduled / Recorded) • Author - Group Activities tab			

1. Project Overview

Your team is required to develop a **secure e-commerce system** using a **Client–Server Architecture**.

The system must demonstrate:

- Secure API communication
- System integration (Frontend ↔ Backend ↔ External API)
- Data conversion (JSON → CSV/XML)

2. System Architecture

You must build:

- Backend (API Server)
 - Handles business logic & security
 - Provides REST API endpoints
- Frontend (Web Client)
 - Calls backend using HTTP
 - Displays data to users

***Both MUST be fully separated (decoupled)**

3. Core Functional Requirements

- Authentication & Security (MANDATORY)
 - JWT-based login system
 - Password hashing (bcrypt or equivalent)
 - Role-Based Access Control (RBAC): Customer / Administrator
- Core API (CRUD)
- Integration System
 - External API Integration
 1. Fetch data from public API (e.g., weather API)
 2. Convert response
 - Payment Webhook Simulation
 1. Simulate payment gateway callback
 - Data Conversion

1. Retrieve order data (JSON)
 2. Convert to: CSV or XML
 3. Allow download
- d. Frontend Requirements
- i. Authentication Pages
 - ii. Customer Dashboard
 - iii. Admin Panel

NOTES:

- Label all figures (screenshot) with: Figure number and title
- Use only the given cover format.
- Submit your report to your lecturer for the lab session.
- Please attach the Evaluation Form to the report.
- Fill up group members' names and IDs in the Evaluation Form.
- Submit 1 FOLDER containing:
 1. Full report.
 2. One compiled, recorded demo video.
 3. Source Code, but not including node modules.
 4. Database file.



UNIVERSITI TUN HUSSEIN ONN MALAYSIA

PROJECT REPORT:
Secure-by-Design E-Commerce Platform
(Client–Server Architecture)

SEM II 2025/2026

COURSE NAME	:	INTEGRATED SYSTEM TECHNOLOGIES
COURSE CODE	:	BIT 22103
SECTION NO	:	1 / 2 / 3
LECTURER	:	
STUDENT NAME & MATRIC NO	:	1. XXXXXXXX (AI XXXXX) 2. YYYYYYYY (CI YYYYY) 3. ZZZZZZZZ () 4. AAAAAAA ()

COGNITIVE: __ /5%; PSYCHOMOTOR: __ / 10%; AFFECTIVE: __ /5%

EVALUATION FORM:

NO.	NAME	MATRIC NUMBER	TOTAL MARKS (A + B + C)
1.			
2.			
3.			
4.			

A. Documentation – Project Report (COGNITIVE)

Criteria	Rating	Marks
Explain the problem context and system requirements for a secure e-commerce platform (Client–Server, API, Integration). [C2]	① ② ③ ④ ⑤	
Clearly explain system design including input/output flow, API requests/responses, and basic logic (if-else, validation, data handling). [C4]	① ② ③ ④ ⑤	
Analyze key components: authentication (JWT), RBAC, API integration, and data conversion. [C4]	① ② ③ ④ ⑤	
Propose and justify secure and effective solutions (e.g., rate limiting, hashing, API protection). [C4]	① ② ③ ④ ⑤	
Proper references and formatting according to UTHM thesis/report guidelines. [C2]	① ② ③ ④ ⑤	
Total		/25 *5 =

B. Technical Skills (PSYCHOMOTOR)

Criteria	Rating	Marks
Execute: Successfully set up Backend API server and database (authentication & CRUD). [P3]	① ② ③ ④ ⑤	
Practice: Successfully implement secure API (JWT, RBAC, validation, rate limiting). [P4]	① ② ③ ④ ⑤	
- Practice: Successfully implement integration features: [P4] - External API - Payment webhook - Data export (JSON → CSV/XML)	① ② ③ ④ ⑤	
Demonstrate: Complete system workflow (Frontend ↔ Backend ↔ API integration). [P3]	① ② ③ ④ ⑤	
Total		/20 *10 =

C. Presentation (AFFECTIVE)

Criteria	Rating	Marks
Organize and deliver presentation within allocated time. [A1]	① ② ③ ④ ⑤	
Demonstrate clear, confident explanation of system architecture and features. [A2]	① ② ③ ④ ⑤	
Present key functionalities (security, API integration, data conversion) effectively. [A4]	① ② ③ ④ ⑤	
Respond to questions accurately and confidently (technical & security aspects). [A3]	① ② ③ ④ ⑤	
Autonomy & Responsibility (Teamwork): Participation and collaboration among group members. [A3] 1 (Very weak) – Not able to work in a team 2 (Weak) – Poor ability of : Teamwork; Collaboration in reaching consensus on an issue 3 (Fair) – Satisfactory ability of : Teamwork; Collaboration in reaching consensus on an issue 4 (Good) – Good ability of : Teamwork; Collaboration in reaching consensus on an issue 5 (Very good) – Excellent ability of : Teamwork; Collaboration in reaching consensus on an issue	① ② ③ ④ ⑤	
Total		/25 *5 =